



TECHNICAL MAINTENANCE MANUAL

Water Tank Maintenance Manual - UH-TANK-703

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Client	UrbanHive Facility Technologies Pvt. Ltd.
Site	GreenPark Meridian Residency, Whitefield, Bengaluru
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Review cycle	Annual or after any major incident

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1. Equipment Overview and Technical Specifications

Overhead RCC domestic water storage tank with level instrumentation, overflow alarm, chlorination dosing, and structural health monitoring at GreenPark Meridian header level.

Model code: UH-TANK-703

Asset registry: Record serial number and installation tower in AMIS before first energization.

Electrical supply: Level sensors 24VDC, dosing pump 230V, telemetry to BMS via 4-20mA transmitter

Environmental limits: Ambient 0-45°C operational; store spare parts below 35°C and 60% RH.

Standards: UrbanHive engineering bulletin UH-ENG-AMIS-2024; applicable Indian Standards as cited per asset class.

Documentation package: This manual, wiring diagram sheet WD-UH-TANK-703, commissioning checklist CC-045, and spare parts catalogue SP-UH-TANK-703.

Technicians must verify nameplate data matches AMIS asset record UH-TANK-703 before applying procedures in this document. Discrepancies must be logged as a data quality ticket to UrbanHive Data Engineering within one business day.

2. Installation Guidelines

Waterproofing membrane inspection before commissioning. Level float set at 90% high, 20% low. Overflow discharge routed to storm drain with mosquito netting.

Pre-commissioning checks:

- Verify incoming power quality: voltage balance within 2%, THD < 5% on dedicated feeder.
- Confirm earthing resistance <= 1 ohm at equipment location; bond to society main earth bar.
- Inspect shipping damage; photograph defects for vendor warranty within 48 hours.
- Complete mechanical alignment, torque logs, and insulation resistance tests per UrbanHive QC form UH-QC-INST-02.

Post-installation: Upload signed commissioning certificate, as-built photos, and warranty start date to AMIS. Tag physical asset with QR code linking to this manual revision v1.0.0.

3. Preventive Maintenance Schedule

UrbanHive mandates the following intervals for GreenPark Meridian. Adjust only with written approval from Head of Maintenance.

- Daily: level trend vs pump run hours.
- Weekly: overflow path clear; ladder access secure.
- Monthly: chlorine residual test 0.2-0.5 ppm at farthest tap.
- Quarterly: internal cleaning if turbidity spike; inspect cathodic protection anodes.
- Annual: structural crack survey; ultrasonic thickness spot check on steel fittings.

Documentation: Each PM visit must generate a service log in AMIS with parts consumed, meter readings, and pass/fail checklist. Missed PM windows trigger automatic escalation to Facility Manager after 72 hours.

4. Troubleshooting Guide

Use isolation procedure: observe symptom ? confirm with meter/test ? apply remediation ? functional test ? document in AMIS.

If fault persists after two visits, escalate to UrbanHive vendor support with fault code and photos.

Fault 01 - Code T101

Symptom: Level sensor drift vs manual dip

Remediation: Calibrate 4-20mA transmitter; clean float guide tube.

Fault 02 - Code T205

Symptom: Overflow alarm during night fill

Remediation: Adjust fill solenoid close time; verify high float NC contact.

Fault 03 - Code T312

Symptom: Low chlorine residual at tap

Remediation: Inspect dosing pump stroke; prime chemical line; test ORP sensor.

Fault 04 - Code T408

Symptom: Dosing pump air lock

Remediation: Bleed injection point; verify tank chemical level; flush with clean water.

Fault 05 - Code T501

Symptom: BMS shows empty while tank 60%

Remediation: Replace ultrasonic sensor; shield from direct sun on rooftop.

Fault 06 - Code T602

Symptom: Sediment discharge at apartments

Remediation: Schedule tank desludging; flush rising main; inspect inlet screen.

Fault 07 - Code T701

Symptom: Leak at manhole frame

Remediation: Re-grout frame; replace gasket; pressure test after 24h cure.

Fault 08 - Code T803

Symptom: Cathodic protection alarm

Remediation: Replace sacrificial anode; measure potential -850mV vs Cu/CuSO₄.

Fault 09 - Code T902

Symptom: Ladder safety cage damage

Remediation: Barricade access; weld repair per structural engineer approval.

Fault 10 - Code T1001

Symptom: Header pressure instability linked to tank

Remediation: Coordinate with booster pump PID; verify non-return valve on fill line.

Fault 11 - Code T1103

Symptom: Mosquito breeding complaint near overflow

Remediation: Install netting; add oil layer protocol per BBMP advisory.

Fault 12 - Code T1202

Symptom: Telemetry loss to AMIS

Remediation: Test GSM/RS485 converter; verify SIM data plan active on rooftop RTU.

5. Spare Parts List

Maintain minimum stock at GreenPark Meridian central store (Basement B1, Room ST-04).

Part No.	Description	Min stock
SP-T01	Level float switch	3
SP-T02	Dosing pump tube kit	5
SP-T03	Chlorine test reagent	10
SP-T04	Ultrasonic level sensor	2
SP-T05	Manhole gasket EPDM	4
SP-T06	Sacrificial anode Mg	6

Reorder when stock falls below minimum. Use UrbanHive procurement portal PR-UH-OPS for approved vendors only.

6. Safety Warnings and Compliance

Confined space entry permit mandatory inside tank. Two-person rule. Gas test O2 > 19.5% before entry. Lock-out fill line during internal work. UrbanHive EHS notification 24h prior.

Permit requirements: Hot work, confined space, work-at-height, and electrical isolation require digital permit in AMIS before start.

PPE minimum: Safety shoes IS 15298, helmet IS 2925, eye protection IS 5983, gloves appropriate to task.

Incident reporting: Any injury, near-miss, or property damage - complete IR-01 within 30 minutes; notify security desk ext. 9911.

7. Manufacturer Contact and UrbanHive Support

OEM technical support: +91-80-5555-0045 (08:00-20:00 IST)

Email: support-uh-tank-703@urbanhive-vendor.example.com

UrbanHive Ops escalation: maintenance@urbanhiveops.example.com - Priority 1 response within 4 business hours

AMIS helpdesk: Log ticket category `Water Tank` with asset_id from tower registry

Emergency (life safety assets): 24x7 hotline +91-80-5555-0911

Revision history: v1.0.0 - Initial release for GreenPark Meridian AMIS integration and UrbanHive contractor certification programme.

End of document MAN-045